

# DEFENSE COLLEGE

## Remedial Examination for Year I Medical Students

Date: August 31, 2012

Time allowed: 3:00 hours.

Name: \_\_\_\_\_

# HISTOLOGY

**MCQ:** Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer or completion that is best in each case. **(Total : 100 points)**

- \_\_\_\_\_1. Which of the following is the site of **steroid hormone synthesis**?  
a) Golgi complex                      b) mitochondria                      c) lysosomes                      d) SER                      e) RER
- \_\_\_\_\_2. With regard to the **centrioles**, which statement is **false**? **Centrioles:**  
a) are composed of nine sets of microtubules arranged in triplets.  
b) are found at the base of cilia and flagella.  
c) are also involved with cell division.                      d) self-duplicate in S phase of the cell cycle.  
e) are associated with membrane activities such as endocytosis, exocytosis, and cell migratory activities.
- \_\_\_\_\_3. With regard to the **ribosomes**, which statement is **false**? **Ribosomes:**  
a) are RNA - rich cytoplasmic granules.                      b) are the sites of protein synthesis.  
c) are intensely basophilic.                      d) are assembled in the nucleoli.  
e) are membrane bounded organelles and are involved in the breakdown of hydrogen peroxide.
- \_\_\_\_\_4. Which of the following is the longest and most variable **cell-cycle phase** in continuously dividing cells?  
a) G<sub>0</sub>                      b) G<sub>1</sub>                      c) G<sub>2</sub>                      d) M                      e) S
- \_\_\_\_\_5. Concerning the **morphology of a cell**, select the **correct** statement.  
a) rRNA is manufactured in the rough endoplasmic reticulum (RER).  
b) Mitochondria are involved in membrane retrieval, recycling and redistribution.  
c) Lipofuscin is a wear-and-tear pigment, otherwise called age pigment and is found in non-dividing cells.  
d) Microtubules are more stable than microfilaments.                      e) Phospholipids are synthesized by peroxisomes.
- \_\_\_\_\_6. Which of the following contain abundant **basal mitochondria** between infoldings of the basal plasma membrane?  
a) ion-transporting cells                      b) mucus-secreting cells                      c) peptide-secreting cells  
d) protein- secreting cells                      e) steroid secreting cells
- \_\_\_\_\_7. With regard to the **epithelial tissue**, select the **incorrect statement**.  
a) An example of epithelioid tissue is the parenchyma of the adrenal gland.  
b) Endothelium is simple squamous epithelium that lines the vascular system.  
c) Goblet cells are associated with simple columnar epithelium of the gastrointestinal tract.  
d) The lining of the urinary bladder is transitional epithelium.                      e) Serosa lines the respiratory tract.

- 8. Type of **epithelium** that covers the luminal surface of expandable organs is:  
a) Transitional epithelium                      b) Simple cuboidal epithelium                      c) Pseudostratified epithelium  
d) Stratified squamous non keratinized epithelium                      e) Stratified squamous keratinized epithelium
- 9. One of the following structures is lined by **stratified columnar epithelium**.  
a) Urinary bladder.                      b) Conjunctiva.                      c) Esophagus.                      d) Vagina.                      e) Trachea.
- 10. With regard to the **epithelial tissue**, which statement is **true**?  
a) Arteriovenous anastomosis provides nutrition to individual cells.  
b) Desmosomes permit the passage of ions between adjacent cells.  
c) Microvilli contain sheaf of microtubules with a 9+2 axoneme pattern.  
d) The major constituents of epithelia is abundant extracellular matrix between the rows of cells.  
e) Merocrine secretion is characterized by discharge of a secretion product with no loss of cytoplasm or membrane.
- 11. With regard to **holocrine secretion**, which statement is **true**?  
a) it occurs in sebaceous glands.                      b) it does not occur in endocrine glands.  
c) involves little or no loss of cytoplasm.                      d) involves secretion of chlorine or bromine ions.  
e) is the typical mode of secretion of DNES cells.
- 12. **In pseudostratified columnar epithelium,**  
a) all nuclei lie at the same depth from the surface.                      b) all cells border onto the lumen  
c) all cells touch the basal lamina.                      d) cells are always of ectodermal origin.  
e) all cells are ciliated.
- 13. Which of the following cell types produces and maintains all **extracellular matrix components of connective tissue**?  
a) plasma cell                      b) mast cell                      c) mesenchymal cell                      d) macrophage                      e) fibroblast.
- 14. Which of the following cells characteristically **synthesize and secrete immunoglobulins**?  
a) fibroblasts                      b) Histiocyte                      c) plasma cells                      d) macrophages                      e) mast cells
- 15. **Mast cells do not secrete**  
a) histamine                      b) heparin                      c) SRS-A (slow reacting substance of anaphylaxis)  
d) catalase                      e) ECF-A (eosinophilic chemotactic factor of anaphylaxis)
- 16. **Elastic cartilage,**  
a) is found in the annulus fibrosus of the intervertebral disks.  
b) is the primary skeletal tissue in the fetus.                      c) has no identifiable perichondrium.  
d) the collagen in its matrix is mainly type II.                      e) is the most widely distributed cartilage type in the body.
- 17. Which of the following cells belong to the **mononucleated phagocytic system** in the **epidermis**?  
a) Merckel's cells                      b) Dust cells                      c) Langerhan's cells                      d) Kupffer cells                      e) Cells of Ito
- 18. Which of the following statements is **correct** regarding the **bone**?  
a) Organic portion of a bone matrix consists primarily type III collagen.  
b) Sharpey's fibers are calcified collagen bundles that run in concentric manner.  
c) Osteoclasts have processes that run in their canaliculi.  
d) Osteoblasts are found as stratified squamous epithelium around the endosteum.  
e) Contains matrix with more inorganic components than organic component in its dry weight.
- 19. Which of the following cell types in **bone** is derived from **blood monocytes**?  
a) endothelium                      b) fibroblast                      c) osteoblast                      d) osteoclast                      e) osteocyte.

- 20. **Primary bone**,  
a) has a higher mineral content than the secondary bone.  
b) may appear as either spongy or compact bone.  
c) is characterized by the presence of Volkmann's canal.  
d) is also termed lamellar bone.  
e) is the first bone tissue to appear during both intramembranous and enchondral bone formation.
- 21. **Compact (dense) bone**,  
a) is the predominant bone tissue in the epiphyses of adult long bones.  
b) is characterized by the presence of osteons.  
c) is also called cancellous bone.  
d) can be either primary or secondary bone.  
e) is typically more radiolucent than other types of bones.
- 22. What type of **adipose tissue** tends to increase as humans age?  
a. Brown adipose tissue      b. White adipose tissue      c. Unilocular adipose tissue  
d. Multilocular adipose tissue      e. Both b and c
- 23. Which of the following cell types is found predominantly in **white matter**?  
a) fibrous astrocytes    b) microglia    c) satellite cells    d) protoplasmic astrocytes    e) Schwann cells
- 24. With regard to the **dendrites**,  
a) each has a constant diameter along its entire length.  
b) may contain Nissl bodies.  
c) each ends with a typical terminal bouton.  
d) are typically myelinated.  
e) contain neurosecretory vesicles.
- 25. Which of the following cell types may provide **myelin** for several axons simultaneously?  
a) ependymal cells    b) oligodendrocytes    c) microglia    d) Schwann cells    e) satellite cells
- 26. Which of the following consists mainly of **axons and neuroglial cells**?  
a) arachnoid mater    b) dura mater    c) white matter    d) gray matter    e) pia mater
- 27. Which of the following statements is **not true** regarding the **supporting cells in the nervous tissue**:  
a) Oligodendrocytes are found in both gray and white matters of the central nervous system.  
b) Satellite cells surround the perikarya of the spinal ganglia in a stratified cuboidal fashion.  
c) Microglia are the smallest in size among the neuroglial cells.  
d) Astrocytes serve in conducting metabolites between blood vessels and neurons.  
e) Ependymal cells line both the ventricles of the brain as well as the central canal of the spinal cord.
- 28. Which of the following tissue types **does not** include **secretion** as a major function of its constituent cells?  
a) connective tissue    b) epithelial tissue    c) muscle tissue    d) nerve tissue    e) all of them
- 29. **Pericytes are**  
a) *specialized cardiac muscle cells*  
b) are attached to the outside of capillaries  
c) *specialized smooth muscle cells*  
d) *terminally differentiated*  
e) *multinucleated*
- 30. **Which of the following vessels is likely to contain the most vasa vasorum in its tunica media?**  
a) *aorta*    b) *arteriole*    c) vena cava    d) *thoracic duct*    e) *venule*
- 31. **Blood capillaries,**  
a) *contain valves*    b) *are blind-ended tubes*    c) *are lined by cells derived from endoderm*

d) characteristically lack occluding junctions between their lining cells  
lymphocytes

e) carry

—32. A differential cell count of a blood smear from a patient with a parasitic infection is likely to reveal

an increase in the circulating numbers of which of the following cell types?

- a) basophils      b) eosinophils      c) neutrophils      d) monocytes      e)

platelets

—33. Which of the following statements is false with regard to the histology of the heart?

- a. Adipose tissue that generally surrounds the heart accumulates in epicardium.  
b. Sinoatrial node consists of double layers of endocardium anchored to fibrous tissues.  
c. Coronary arteries branch to supply the epicardium, myocardium, and endocardium.  
d. Endocardium is the thickest in atria and the thinnest in ventricles, especially the left ventricle.  
e. Cardiac muscle is more vascularized and has more abundant mitochondria than does skeletal muscle.

—34. Which of the following cell types is capable of returning to the circulation after leaving the blood

to enter the connective tissue?

- a) basophils      b) eosinophils      c) lymphocytes      d) monocytes      e)

platelets

—35. Which of the following substances is present in higher concentrations in plasma than in serum?

- a) albumin      b) fibrinogen      c) immunoglobulin      d) glucose      e)

serotonin

—36. Which of the following immunoglobulin types is most important in conferring passive immunity on

the fetus and on the newborn infant?

- a) IgA      b) IgD      c) IgE      d) IgG

e) IgM

—37. Which of the following is partly encapsulated and covered by nonkeratinized stratified squamous epithelium?

- a) appendix      b) lingual tonsil      c) palatine

tonsil

- d) Peyer's patches      e) pharyngeal tonsil

—38. Which of the following lymphoid organs receives afferent lymphatic vessels?

- a) appendix      b) palatine tonsil      c) spleen      d) thymus      e)

lymph node

—39. Which of the following is not a thymic dependent secondary lymphoid tissue?

- a) Periarterial lymphoid sheath of spleen      b) Paracortical zone of the lymph

node

- c) Peyer's patches of the ileum      d) Billroth's cords of the spleen      e) None of the above

—40. Regarding the thymus, which statement is true?

- a) Has blood-thymus barrier in the cortex.

- b) Is relatively more developed with

increasing age.

- c) Has Hassall's corpuscles located in the cortex.

- d) Has lymphoid nodules in

its cortex.

**e) Contains more B lymphocytes than the T lymphocytes.**

—41. The **spleen**:

- a) Has smooth myocytes in its capsule.
- b) Filters lymph.
- c) Has no B lymphocytes.
- d) Has an outer cortex and inner medulla.
- e) Contains periarterial lymphoid sheath in its red pulp.

—42. Which of the following is lined by a **mucosa**?

- a. Peritoneal cavity
- b. Pericardial cavity
- c. Pleural cavity
- d. Alimentary canal
- e. All of the above are lined by a mucosa

—43. Which of the following types of **lingual papillae** lacks taste buds?

- a) circumvallate
- b) filiform
- c) foliate
- d) fungiform
- e) none

—44. Regarding the **stomach**:

- a) Has its muscularis externa made up of skeletal muscle fibres.
- b) Has its gastric glands best developed in the pylorus than the other regions.
- c) Has chief cells that are intensely eosinophilic
- d) Has parietal cells in its glands that secrete HCl.
- e) Is able to withstand the acidic environment because it is lined by stratified squamous keratinized epithelium.

—45. Concerning the **teeth and associated tissues**, which statement is **true**?

- a) Dentin is more calcified than enamel.
- b) Pulp is made up of dense connective tissue.
- c) Enamel is replaced in the root by cementum.
- d) Odontoblasts make enamel.
- e) Periodontal membrane is loose connective tissue.

—46. Which of the following statements is **correct** regarding the **intestine**?

- a) Large intestine possesses branched alveolar glands.
- b) The striated border of the small intestine is made by microvilli.
- c) Small intestine has taenia coli.
- d) Large intestine contains villi.
- e) None of the above.

—47. **Mucosal glands of the pylorus**,

- a) resemble those in the body of the stomach.
- b) secrete more pepsinogen than do fundic glands.
- c) have shallower pits than those in the fundic region.
- d) secrete mainly mucus.
- e) contain abundant parietal cells in their basal half.

—48. **Permanent circular folds (plicae circulares)** are characteristic structural feature of the walls of which of the following organs?

- a) anal canal
- b) colon
- c) cecum
- d) small intestine
- e) stomach

—49. Which of the following organs characteristically contains the most **goblet cells** in the mucosa covering its villi?

- a) duodenum
- b) ileum
- c) colon
- d) cecum
- e) stomach

—50. **Kupffer cells**,

- a) are nonphagocytic
- b) contain lysosomes
- c) are derived from circulating neutrophils
- d) resemble hepatocytes in HE-stained sections
- e) border on the bile canaliculi

—51. Which of the following glands lacks **serous acini but contains serous demilunes**?

- a) liver
- b) parotid gland
- c) pancreas
- d) sublingual gland
- e) submandibular gland

—52. The **absorptive cells of the small intestine**,

- a) are also called enteroendocrine cells.
- b) many microvilli cover their basal surfaces.
- c) absorb lipids by active transport.
- d) use absorbed lipids to synthesize triglycerides.
- e) undergo mitosis at the tips of the villi and are sloughed into the crypts.

53. **Brunner's glands**,  
a) are characteristic components of the jejunal wall. b) lie in the submucosal layer.  
c) empty their secretions into the lacteals. d) are composed of collections of goblet cells.  
e) produce serous secretion that is rich in digestive enzymes.
54. Which of the following components of the respiratory tract wall increase(s) in amount from trachea to alveoli?  
a) cilia goblet cells b) elastic fibers c) smooth muscle d) cartilage e)
55. Which of the following structures is (are) partly lined by nonkeratinized stratified squamous epithelium?  
a) nasopharynx bronchioles b) larynx c) bronchi d) trachea e)
56. Pulmonary surfactant,  
a) prevents alveolar collapse by increasing surface tension.  
b) consists of a monolayer of phospholipids overlying an aqueous hypophase.  
c) excessive production in the lungs of premature infants causes respiratory distress syndrome.  
d) is secreted by type I pneumocytes. e) may promote the growth of bacteria.
57. Bronchioles:  
a) Have well developed hyaline and elastic cartilages b) Are lined by stratified squamous epithelium  
c) Have no muscle d) Have Clara cells e) Have mucous producing glands
58. C-shaped hyaline cartilages are characteristic features of:  
a) Larynx. b) Trachea. c) Bronchi. d) Bronchioles. e) Alveolar ducts
59. Which of the skin's many functions are partly or completely provided by components of the dermis?  
a) protection from dehydration b) regulation of blood pressure  
c) protection from ultraviolet radiation d) synthesis of keratin e) synthesis of melanin
60. Which of the following epidermal layers is most likely to contain both **desmosomes** & **hemidesmosomes**?  
a) stratum basale b) stratum corneum c) stratum granulosum  
d) stratum lucidum e) stratum spinosum
61. Which of the following is (are) often absent in thick skin but typically present in thin skin?  
a) arrector pili muscles b) Meissner's corpuscles c) sweat glands  
d) stratum basale e) stratum spinosum
62. Which of the following is the modified smooth muscle cell responsible for the secretion of renin?  
a) juxtaglomerular cell b) mesangial cell c) podocyte  
d) polkissen (extraglomerular mesangial cell) e) macula densa cells
63. What are the **ducts of Bellini**?  
a. Collecting tubules b. Distal convoluted tubules c. Proximal convoluted tubules  
d. Loops of Henle e. Medullary rays
64. Which of the following cell types comprises the visceral layer of Bowman's capsule?

- a) endothelial cell
- b) juxtaglomerular cell
- c) mesangial cell
- d) podocyte
- e) polkissen (extraglomerular mesangial cell)

- 65. Which of the following is characteristic of the **proximal** but **not** of the **distal convoluted tubules**?
- a) convoluted portion lies in the renal cortex
  - b) epithelial lining is similar to that of the loop of Henle
  - c) exhibits a brush border
  - d) lining cells are basophilic
  - e) secretes glucose and amino acids into the filtrate
- 66. Which of the following is **not** a **function of the kidney**?
- a. Erythropoietin production
  - b. Vitamin D modification
  - c. Acid base balance
  - d. Aldosterone production
  - e. Renin production
- 67. Which of the following is **not** the function of **mesangial cells**?
- a. synthesis of extracellular matrix.
  - b. endocytosis and contractility.
  - c. maintenance of acid-base balance in the blood.
  - d. structural support to the glomerulus.
  - e. increase blood flow and the effective surface area available for filtration.
- 68. In the **urinary system**, which statement is **false**?
- a. The ureter is lined by transitional epithelium and it does not contain lymphatics.
  - b. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
  - c. Distal convoluted tubule is longer and more convoluted than proximal convoluted tubule.
  - d. Calyces are lined by transitional epithelium.
  - e. Aldosterone acts on the distal convoluted duct.
- 69. Which of the following hormones is carried to its target(s) by the **hypophyseal portal system**?
- a) ADH
  - b) ACTH
  - c) FSH
  - d) TRH
  - e) TSH
- 70. In the **urinary system**, which statement is **true**?
- a. Interstitial cells in the renal medulla secrete erythropoietin.
  - b. Macula densa are located in the urinary pole of the nephrons.
  - c. The male urethra contains transitional epithelium throughout its length.
  - d. Endothelial cells of glomerular capillaries are of non-fenestrated variety.
  - e. The distal convoluted tubules possess neither brush borders nor apical canaliculi.
- 71. Which of the following hormones is **most** responsible for **increasing osteoclast number and activity**?
- a) ADH
  - b) calcitonin
  - c) estrogen
  - d) parathyroid hormone
  - e) testosterone
- 72. **Ventricles of the heart,**
- a) **receive blood directly from the venae cavae and the pulmonary veins.**
  - b) **their walls contain the right and left bundle branches.**
  - c) **contain more abundant elastic fibers than do the atria.**
  - d) **are located at the base of the heart.**
  - e) **their myocardial cells contain abundant granules.**
- 73. **Tunica intima,**
- a) **contains some small capillaries.**
  - b) **is homologous to the pericardium of the heart.**
  - c) **includes a layer of simple squamous epithelium.**
  - d) **is a fenestrated sheet of elastin.**
  - e) **is separated from tunica media of arteries by the external elastic lamina.**
- 74. Which of the following statements is false about the circulatory system?
- a) **Lymphatic capillaries can penetrate the media of veins.**
  - b) **Vasa vasorum are more frequent in veins than in arteries.**
  - c) **The tunica adventitia tends to be much larger in veins than in arteries.**

- d) *The concentration of glycosaminoglycans is higher in arterial than in venous tissue.*  
e) Sinusoids of the liver and the glomeruli of the kidney typically possess continuous endothelium.

———75. Continuous capillaries,

- a) *possess unusually wide lumina.* b) *have abundant fenestrations.*  
c) *typically follow a tortuous (twisting) course.* d) are common in brain and muscle tissue.  
e) *often have phagocytic cells inserted between the endothelial cells of their lining.*

———76. *Which of the following statements is false with regard to the circulatory system?*

- a. *Post capillary venules lack a smooth muscle media.*  
b. *Carotid bodies are chemoreceptors sensitive to hypoxia.*  
c. *The brachiocephalic and subclavian arteries are elastic arteries.*  
d. Weibel-Palade granules are found only in endothelial cells of capillaries.  
e. *Venous valves of the lower limbs have two semilunar folds that are lined on both sides by endothelium.*

———77. Concerning the **chromophobes** of **pars distalis of adenohypophysis**, which statement is **false**?

- a. may represent stem cells that have not yet differentiated into hormone-producing cells.  
b. maintain favorable fluid and electrolyte environment for the chromophil secretory cells.  
c. may be acidophils or basophils that are degranulated and thereby are depleted of hormone.  
d. do not stain intensely. e. may have minimal or no hormonal content.

———78. Which statement is **false** with regard to the **epiphysis cerebri {pineal body}**?

- a. Melatonin may "delay" puberty through antigonadotropic effects.  
b. Its secretory activity is stimulated by darkness and inhibited by light.  
c. brain sand or corpora arenacea increase in size and number with increasing age.  
d. is surrounded by pia mater, which functions as its capsule. is covered by pia mater.  
e. None of the above.

———79. Which statement is **true** with regard to the **endocrine system**?

- a. Myxedema is considered to be an autoimmune disease caused by hyperthyroidism.  
b. Calcitonin, secreted by the parathyroid gland, stimulates calcium resorption in the bones.  
c. The small and most numerous oxyphil cells of the parathyroid glands stain strongly basophilic.  
d. Both zona fasciculata and zona reticularis of the suprarenal gland depend on ACTH to sustain their function and survival.  
e. The cells of the zona fasciculata of the suprarenal gland are called chromaffin cells because the granules of these cells can be stained strongly of the suprarenal gland with potassium bichromate.

———80. The **neurohypophysis**,

- a) contains chromophils. b) includes the pars intermedia. c) secretes ACTH.  
d) contains axons of hypothalamic neurons. e) hormone production is regulated by releasing factors.

———81. **Arrektor pili muscles**,

- a) are composed of bundles of myoepithelial cells. b) help push sebum onto the skin surface.  
c) have no effect on hair position in humans. d) contain capillary networks that nourish growing hairs.  
e) attach directly to the external root sheath by desmosomes.

———82. What vessels arise from the **efferent arteriole**?

- a. Afferent arteriole b. Efferent arteriole c. Capillary tuft d. Peritubular capillaries e. Vasa recta

———83. **Testis** has the following coverings **except**:

- a. tunica adventitia b. tunica vasculosa c. tunica albuginea d. tunica vaginalis e. None

- 84. In the **testis**, which statement is **true**?
- Interstitial cells (of Leydig) produce testicular fluid.
  - The seminiferous tubules are lined with a complex stratified epithelium.
  - The secondary spermatocytes are the largest cells of the spermatogenic lineage.
  - The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
  - During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.
- 85. Which of the following is **NOT** considered part of the **male genital duct system**?
- Rete testis
  - Tubuli recti
  - Seminal vesicles
  - Ductus deferens
  - Ductus epididymidis
- 86. Which statement is **false** with regard to the **prostate gland**?
- It is the largest accessory structure of the male reproductive system.
  - The glands of the prostate lie in a fibromuscular stroma, also referred to as a fibroelastic stroma.
  - Corpora amylacea {prostatic concretions} are the spherical structures seen in some prostatic alveoli.
  - It is composed of compound tubuloalveolar glands.
  - It produces testicular androgens.
- 87. Which of the following statements is **false** in the **male reproductive system**?
- The secretion of seminal vesicle is particularly rich in fructose.
  - Secondary spermatocytes are smaller than the primary spermatocytes.
  - The epithelium lining the ductus epididymis is pseudostratified columnar.
  - Tunica albuginea is the thickest in epididymis.
  - The subcutaneous tissue in penis is devoid of fat.
- 88. Which of the following is true of the **ampulla** of the **uterine tube** but **not** of its **isthmus**?
- Contains large mucosal folds that are extensively branched.
  - Contains smooth muscle in its walls.
  - Is covered by serosa derived from the peritoneum.
  - Contains ciliated epithelial cells.
  - Contains mucus-secreting peg cells.
- 89. The **cornea**,
- constitutes the anterior half of the tunica fibrosa.
  - is composed of 10 layers.
  - is avascular.
  - accounts for eye color.
  - is attached to the ciliary body by the zonule.
- 90. Which of the following statements is **true** about the **female reproductive system**?
- Mucosa of the uterus is commonly called perimetrium.
  - Fornix of vagina is lined by stratified squamous epithelium.
  - The life span of corpus luteum of pregnancy is about 10 lunar months.
  - Primary oocyte completes its first meiotic division at the time of fertilization.
  - Nonciliated secretory cells present in the lining epithelium of uterine tube synthesise and store glycogen.
- 91. Which of the following statements is **true** in the **female reproductive system**?
- The cells of theca interna secrete progesterone.
  - Germinal (surface) epithelium covering the ovary gives rise to ova.
  - Glassy membrane is the last structure to disappear in an atretic follicle.
  - The alkaline environment of vagina prevents growth of microorganisms.
  - The basic pH in the vaginal lumen is due to the secretion of HCl by the tubuloalveolar glands of the vaginal walls.
- 92. In the **eye**, one of the following statements is **false**.
- The ciliary body generates the aqueous humor.
  - The lens does not contain blood vessels and nerves.
  - The choroid is the vascular, pigmented layer of the eye.
  - Lacrimal canaliculi are lined by a stratified squamous epithelium.
  - The cornea is covered by an incomplete layer of pigmented cells and fibroblasts on its anterior surface.
- 93. Concerning the histology of the **ear**, one of the following statements is **true**.
- Tectorial membrane is secreted by the interdental cells of the stria vascularis.

- b) The middle ear contains modiolus, cristae ampullaris and semicircular canals.
- c) Pinna consists of an irregular piece of elastic cartilage surrounded by a thick perichondrium.
- d) The lining epithelium of the outer aspect of the tympanic membrane is ciliated columnar interspersed with ceruminous glands.
- e. None of the above.

- \_\_\_\_\_ 94. Which one of these cells is not a cell type routinely found in **loose connective tissue**?  
 a. Fibroblast      b. Microglia      c. Histocyte      d. Plasma cell      e. Mast cell
- \_\_\_\_\_ 95. What type of tissue is a **tendon** composed of?  
 a. Mucous connective tissue      b. Mesenchyme      c. Loose irregular connective tissue  
 d. Dense irregular connective tissue      e. Dense regular connective tissue
- \_\_\_\_\_ 96. What do you call the space where a **chondrocyte** sits in?  
 a. Space of Disse      b. Space of Moll      c. Vacuole      d. Lacuna      e. Howship's Lacuna
- \_\_\_\_\_ 97. Which type of cartilage forms the **hammer, anvil and stirrup (malleus, incus and stapes)**?  
 a. Hyaline cartilage      b. Elastic cartilage      c. Fibrocartilage      d. All of the above      e. None of the above
- \_\_\_\_\_ 98. Select the **incorrect** statement about **hyaline cartilage**.  
 a. It forms the skeleton of the fetus.      b. It forms a template of the bones      c. It forms the intervertebral disk.  
 d. It is characterized by a glassy matrix.      e. The thyroid cartilage is composed of hyaline cartilage.
- \_\_\_\_\_ 99. Select the **incorrect** statement about the **neurons. Neurons:**  
 a. are specialized for conduction.      b. have an extremely long longevity, lasting a lifetime.  
 c. have a high metabolic rate.      d. are unable to survive long without oxygen.      e. None
- \_\_\_\_\_ 100. In the nervous system, which statement is false?  
 a. Satellite cells are found in the ganglia of the peripheral nervous system.      b. Olfactory cell is a bipolar neuron.  
 c. Cerebrospinal fluid is found in the subarachnoid space.      d. The primary component of myelin is lipid.      e. None

# EMBRYOLOGY

**MCQ:**      **Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer or completion that is best in each case.**  
**(Total : 50 points)**

- \_\_\_\_\_ 1. Most **birth defects** occur during which of the following periods/stages of the intrauterine life?  
 a) First 2 weeks      b) Embryonic period      c) Fetal period      d) Last trimester      e) Zygote
- \_\_\_\_\_ 2. The **central tendon of the diaphragm** develops from:  
 a) Dorsal mesentery of the esophagus      b) Pleuroperitoneal membranes  
 c) Pleuropericardial membranes.      d) Lateral body wall      e) Septum transversum.
- \_\_\_\_\_ 3. Structure that is **not** developed in the **third branchial arch** include:  
 a) Lesser cornu of hyoid bone      b) Body of the hyoid bone  
 c) Glossopharyngeal nerve      d) Common carotid artery      e) Stylopharyngeus muscle
- \_\_\_\_\_ 4. Which of the following structures develop from **endoderm of the second pharyngeal pouch**?  
 a) Thymus gland      b) Palatine tonsil      c) Thyroid gland  
 d) Superior parathyroid gland      e) Parafoallicular cells of thyroid gland

- \_\_\_\_\_ 5. The **anterior 2/3<sup>rd</sup> of the tongue** develops at which pharyngeal arch level?  
a) 1<sup>st</sup>                      b) 2<sup>nd</sup>                      c) 3<sup>rd</sup>                      d) 4<sup>th</sup>                      e) 6<sup>th</sup>
- \_\_\_\_\_ 6. Which of the following **salivary glands** develop from ectoderm?  
a) Parotid                      b) Submandibular                      c) Sublingual                      d) Lingual                      e) All of the above
- \_\_\_\_\_ 7. *Secretion of surfactant, which is an important substance for survival of premature infants, appears during which of the following periods of lung development?*  
a) Canalicular.                      b) Bronchiolar.                      c) Terminal sac period                      d) Pseudoglandular                      e) Alveolar
- \_\_\_\_\_ 8. All of the following **appendages of the skin** develop from ectoderm, **except**:  
a) Hairs.                      b) Sweat glands.                      c) Sebaceous glands.                      d) Arrector pili muscle.                      e) Nails.
- \_\_\_\_\_ 9. With regard to **fertilization**, which statement is **false**? **Fertilization**:  
a) restores the normal diploid number of chromosomes in the zygote.  
b) results in variation of human species through mingling of maternal and paternal chromosomes.  
c) causes metabolic activation of the ovum and initiates cleavage of the zygote.  
d) determines chromosomal sex of the embryo.                      e) None of the above.
- \_\_\_\_\_ 10. During the **first week of development**,  
a) the embryo is a solid ball of blastomeres and is called gastrula.  
b) mesoderm is formed which will give rise later to the placental membranes.  
c) the blastocyst is superficially implanted in the uterine wall (endometrium).  
d) developmental disturbances may give rise to major congenital anomalies.  
e) amniotic cavity is developed within the yolk sac as a space between the cytotrophoblast and the embryoblast.
- \_\_\_\_\_ 11. **Neuroepithelial cells of the neural tube** give rise to all of the following cells of the brain **except**:  
a) Microglia.                      b) Astrocytes.                      c) Oligodendrocytes.  
d) Purkinje cells of cerebellar cortex.                      e) Ependymal cells.
- \_\_\_\_\_ 12. During the **third week of development**, which statement is **false**?  
a) somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.  
b) notochord forms the primitive axis of the embryo around which the axial skeleton will form.  
c) the primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.  
d) mesoderm exists between the ectoderm and endoderm everywhere.                      e) neural tube is formed.
- \_\_\_\_\_ 13. Concerning **multiple pregnancies**,  
a) the rate of dizygotic twinning increases with maternal age.  
b) the frequency of monozygotic twinning shows marked racial differences.  
c) twins with one amnion, one chorion and one placenta are always monozygotic.  
d) dizygotic twins always have two amnions and two chorions.                      e) monozygotic twins are of the same sex.
- \_\_\_\_\_ 14. Concerning the **development of the GIT and associated structures**, which statement is **true**?  
a) Liver develops by fusion of ventral and dorsal hepatic buds.  
b) Stomach rotates 270° counterclockwise, similar to the midgut.  
c) The loop of the midgut has the superior mesenteric artery in its axis.  
d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.  
e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.
- \_\_\_\_\_ 15. With regard to the **development of the musculoskeletal system**,  
a) the muscular system develops from neuroectoderm.  
b) the upper limbs rotate laterally through 90° on their longitudinal axes.

- c) premature closure of the frontal (metopic) suture results in scaphocephaly.  
 d) sclerotomes of the somites form the viscerocranium. e) xiphoid process develops during the 6<sup>th</sup> week.

\_\_\_\_\_ **16. Primary brain vesicle** that remains relatively unchanged into the secondary brain vesicle is:

- a) Rhombencephalon                      b) Prosencephalon                      c) Telencephalon  
 d) Mesencephalon                      e) Diencephalons

\_\_\_\_\_ **17. Which of the following componenets plays the most active role in *invading the endometrium* during *blastocyst implantation*?**

- a) epiblast                      b) syncytiotrophoblast                      c) hydroblast  
 d) extraembryonic somatic mesoderm                      e) extraembryonic visceral mesoderm

\_\_\_\_\_ **18. The *prochordal plate* marks the site of the future**

- a) umbilical chord                      b) heart                      c) mouth                      d) anus                      e) nose

\_\_\_\_\_ **19. Which process establishes the *three definitive germ layers*?**

- a) neurulation                      b) gastrulation                      c) craniocaudal folding                      d) lateral folding                      e) angiogenesis

\_\_\_\_\_ **20. The *spiral organ of Corti* is derived from**

- a) both ectoderm and mesoderm                      b) neural crest                      c) endoderm  
 d) mesoderm                      e) surface ectoderm

\_\_\_\_\_ **21. With regard to the *congenital anomalies of the kidneys and ureters* which statement is *false*?**

- a) Accessory renal arteries are end arteries.  
 b) Bilateral renal agenesis is associated with oligohydramnios.  
 c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.  
 d) Unilateral renal agenesis is more common in males than females.                      e) None of the above.

\_\_\_\_\_ **22. Which of the following is not the *adult derivative of the ureteric bud*?**

- a) renal pelvis                      b) major calyx                      c) minor calyx                      d) collecting duct                      e) connecting tubule

\_\_\_\_\_ **23. Which of the following structures in the *eye* is derived from *surface ectoderm*?**

- a) retina                      b) dilator and sphincter pupillae muscles of iris  
 c) corneal endothelium                      d) vitreous body                      e) lens

\_\_\_\_\_ **24. *Urachal cyst* is a remnant of:**

- a) urogenital sinus                      b) urogenital rigde                      c) cloaca                      d) allantois                      e) mesonephric duct

\_\_\_\_\_ **25. The *podocytes of Bowman's capsule* are derived from:**

- a) ectoderm                      b) mesoderm                      c) endoderm                      d) endoderm and mesoderm                      e) neural crest cells

\_\_\_\_\_ **26. A structure found within the adult *female pelvis formed from the gubernaculum* is**

- a) broad ligament                      b) superior ligament of the ovary                      c) medial umbilical ligament  
 d) round ligament of the uterus                      e) median umbilical ligament

\_\_\_\_\_ **27. The *labia minora* arise embryologically from:**

- a) phallus                      b) labioscrotal swellings                      c) sinovaginal bulbs  
 d) urogenital folds                      e) paramesonephric duct

\_\_\_\_\_ **28. The *uterine tubes* of the adult female are derived embryologically from**

- a) mesonephric duct                      b) mesonephric tubules                      c) paramesonephric duct  
 d) paramesonephric tubules                      e) uterovaginal primordium

\_\_\_\_\_ **29. The most common cause of *testicular feminization syndrome* is**

- a) a 46, XO genotype
- b) a 47, XXY genotype
- c) lack of androgen receptors
- d) congenital adrenal hyperplasia
- e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

30. In the male, **failure of urethral folds to fuse completely** results in

- a) hypospadias
- b) epispadias
- c) cryptorchidism
- d) hydrocele
- e) congenital inguinal hernia

31. A **sacroccygeal teratoma** is a tumor that arises from remnants of the

- a) neural plate
- b) cloacal membrane
- c) posterior neuropore
- d) primitive streak
- e) notochord

32. **Intermediate medoderm** gives rise to the

- a) neural tube
- b) heart
- c) kidneys and gonads
- d) somites
- e) notochord

33. **The portal vein is derived from**

- a) vitelline veins
- b) umbilical veins
- c) anterior cardinal veins
- e) posterior cardinal veins
- e) subcardinal veins

34. The **stapedius muscle** that moves the stapes ossicle is innervated by

- a) CN V3
- b) CN XII
- c) CN III
- d) CN VII
- e) cervical nerves C2 and C3

35. **Failure of choroid fissure to close** results in

- a) congenital detached retina
- b) congenital aniridia
- c) congenital aphakia
- d) coloboma iridis
- e) microphthalmos

36. Concerning the **development of the Urinary System**, which statement is **false**?

- a) The kidneys attain their definitive adult position postnatally.
- b) Urinary system begins to develop before the genital system.
- c) The metanephric diverticulum is the primordium of the ureter.
- d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- e) Differentiation of the nephrons depends upon induction by the collecting tubules.

37. With regard to the development of the **endocrine system**, which of the following statements is **false**?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
- b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4<sup>th</sup> pair of pharyngeal pouches.
- c) the thyroid gland is the first endocrine gland to develop in the embryo.
- d) the pituitary gland is ectodermal in origin.
- e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

38. Which of the following is not the adult derivative of the **ureteric bud**?

- a) renal pelvis
- b) major calyx
- c) minor calyx
- d) collecting duct
- e) connecting tubule

39. Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The female urethra is derived from endoderm of the urogenital sinus.
- b) In adults, the urachus is represented by the median umbilical ligament.
- c) The urinary bladder becomes a pelvic organ during the 9<sup>th</sup> week of development.
- d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

40. During the **development of the kidney systems**, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
- b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.

- c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.  
d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated appearance.

**41. During the development of the female reproductive system, which statement is false?**

- a) The urogenital folds remain unfused and form the labia minora.  
b) The cranial portion of the paramesonephric ducts form the uterine tubes.  
c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.  
d) The phallus forms the mons pubis. e) The labioscrotal swellings form the labia majora.

**42. Concerning the development of the genital system, one statement is false?**

- a) Bilateral cryptorchidism may result in sterility.  
b) Hypospadias is constantly associated with exstrophy of urinary bladder.  
c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.  
d) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora.  
e) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as chordee.

**43. With regard to the development of gonads, which statement is false?**

- a) The surface epithelium of the female gonad develop into oogonia.  
b) Sex of an embryo is determined genetically at the time of fertilization.  
c) Germ cells do not appear in the genital ridges until the 6<sup>th</sup> week of development.  
d) By the 8<sup>th</sup> week of gestation, the interstitial cells of Leydig begin to produce testosterone.  
e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

**44. With regard to the vestigial structures derived from embryonic genital ducts, which statement is false?**

- a) The prostatic utericle is homologous to the vagina.  
b) The cranial end of mesonephric duct in males may persist as paradidymis.  
c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis  
d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.  
e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

**45. With regard to development of the eye, which statement is true?**

- a. Failure of choroid fissure to close results in congenital aphakia.  
b. The epithelium of the ciliary body is derived from endoderm.  
c. The optic nerve is a tract of diencephalon that is completely myelinated during the third trimester.  
d. The hyaloid canal (cloquet's canal) is found in the vitreous body.  
e. Aqueous humor is produced by the choroid plexus.

**46. With regard to the development of the ear, which statement is true?**

- a. The spiral organ of Corti is derived from neural crest.  
b. The middle ear cavity develops from the otic vesicle.  
c. The utricular portion of the otic vesicle gives rise to cochlear duct.  
d. Perilymph enters the subarachnoid space via the endolymphatic sac. e. None of the above.

**47. With regard to the development of the head and neck,**

- a) sphenoidal sinuses develop during the early fetal life (10 to 11 weeks)  
b) the critical period of palate development is from the end of the 6<sup>th</sup> week until the beginning of the 9<sup>th</sup> week.  
c) taste buds form on the dorsal surface of the tongue only.  
d) all parenchymal (secretory) tissue of the salivary glands arises by proliferation of the oral epithelium.  
e) by late fetal period, the nasolacrimal duct drains into the inferior meatus in the lateral wall of the nasal cavity.

**48. Concerning the development of the Integumentary system, which statement is true?**

- a) the epidermis is derived from surface ectoderm.

- b) castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.
- c) teeth develop from ectoderm, mesoderm and neural crest cells.
- d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.
- e) toenails and fingernails begin to develop at the tips of the digits postnatally.

**49.** During the development of the digestive system, which statement is **true?**

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the midgut may give rise to the descending colon and sigmoid colon.
- d) the dorsal pancreatic bud forms the uncinat process and part of the head of the pancreas.
- e) the anal membrane normally breaks down by the end of the second trimester.

**50.** Concerning the development of vasculature, which statement is **true?**

- a) distal parts of the 3<sup>rd</sup> pair of aortic arches join the dorsal aortae to form the internal carotid arteries.
- b) the left 4<sup>th</sup> aortic arch becomes the proximal part of the left subclavian artery.
- c) the distal part of the right 6<sup>th</sup> aortic arch becomes the ductus arteriosus.
- d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.
- e) after birth, the proximal parts of the vitelline arteries persist as the superior vesical arteries.

#### **BONUS QUESTIONS.**

1. Typical placenta has a diameter of \_\_\_\_\_ and a thickness of \_\_\_\_\_. It weighs \_\_\_\_\_.
2. Tetralogy of Fallot consists of which cardiac defects?
3. Enumerate the functions [significance] of the amniotic fluid.
4. Enumerate the derivatives of ectoderm.